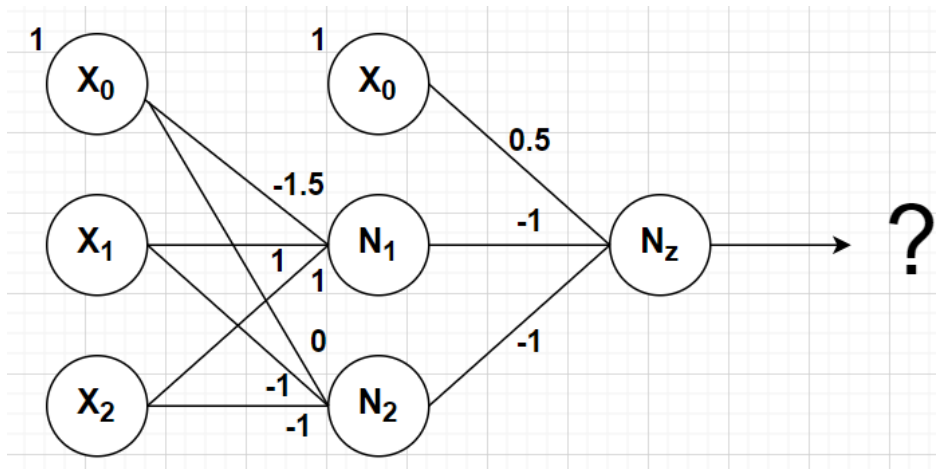


Activity 2 – ECS 171

Question:

What is this gate? Identify the logical gate for every neuron. Show all the steps taken to solve this and identify which logical gate the output of the network corresponds to. Assume the threshold for the activation function is 0.



X1	X2	Y: g(x;w)
0	0	? 0
0	1	? 1
1	0	? 1
1	1	? 0

x_1	x_2	N_1	N_2	N_z
0	0	$1(-1.5) + 0(1) + 0(1) = -1.5 \rightarrow 0$	$0 + 0(-1) + 0(-1) = 0 \rightarrow 1$	$0.5 + 0(-1) + 1(-1) = -0.5 \rightarrow 0$
0	1	$1(-1.5) + 0(1) + 1(1) = -0.5 \rightarrow 0$	$0 + 0(-1) + 1(-1) = -1 \rightarrow 0$	$0.5 + 0(-1) + 0(-1) = 0.5 \rightarrow 1$
1	0	$1(-1.5) + 1(1) + 0(1) = -0.5 \rightarrow 0$	$0 + 1(-1) + 0(-1) = -1 \rightarrow 0$	$0.5 + 0(-1) + 0(-1) = 0.5 \rightarrow 1$
1	1	$1(-1.5) + 1(1) + 1(1) = 0.5 \rightarrow 1$	$0 + 1(-1) + 1(-1) = -2 \rightarrow 0$	$0.5 + 1(-1) + 0(-1) = -0.5 \rightarrow 0$

XOR gate